$\equiv (((\neg p \vee \neg q) \wedge p) \vee ((\neg p \vee \neg q) \wedge q)) \vee ((p \wedge q) \wedge (\neg p \wedge \neg q))$
(2nd Distributive Law)

$\equiv ((p \wedge (\neg p \vee \neg q)) \vee (q \wedge (\neg p \vee \neg q)) \vee (p \wedge (q \wedge (\neg p \wedge \neg q)))$ (2nd Associative Law)

$\equiv ((p \wedge \neg p) \vee (p \wedge \neg q)) \vee ((q \wedge \neg p) \vee (q \wedge \neg q)) \vee (p \wedge (q \wedge (\neg p \wedge \neg q)))$ (2nd Distributive Law)

$\equiv ((p \wedge \neg p) \vee (p \wedge \neg q)) \vee ((q \wedge \neg p) \vee (q \wedge \neg q)) \vee (p \wedge (q \wedge (\neg q \wedge \neg p)))$ (2nd Commutative Law)

$\equiv ((p \wedge \neg p) \vee (p \wedge \neg q)) \vee ((q \wedge \neg p) \vee (q \wedge \neg q)) \vee (p \wedge ((q \wedge \neg q) \wedge \neg p))$ (2nd Associative Law)

$\equiv (F \vee (p \wedge \neg q)) \vee ((q \wedge \neg p) \vee F) \vee (p \wedge (F \wedge \neg p))$ (2nd Negation Law)

$\equiv ((p \wedge \neg q) \vee F) \vee ((q \wedge \neg p) \vee F) \vee (p \wedge (\neg p \wedge F))$ (2nd Commutative Law)

$\equiv ((p \wedge \neg q) \vee (q \wedge \neg p)) \vee (p \wedge (\neg p \wedge F))$ (2nd Identity Law)

$\equiv ((p \wedge \neg q) \vee (q \wedge \neg p)) \vee (p \wedge F)$ (2nd Domination Law)

$\equiv ((p \wedge \neg q) \vee (q \wedge \neg p)) \vee F$ (2nd Domination Law)

$\equiv ((p \wedge \neg q) \vee (q \wedge \neg p))$ (2nd Identity Law)

Definition 2: A formula which is equivalent to a given formula and consists of a product of elementary sums is called a conjunctive normal form of the given formula (CNF).

Example 2: Bring to conjunctive normal form:

1. $(p \rightarrow \neg q) \wedge \neg q$
2. $\neg(p \wedge q) \leftrightarrow (p \vee q)$

1. $(p \rightarrow \neg q) \wedge \neg q \equiv (\neg p \vee q) \wedge \neg q$ (Conjunction-disjunction equivalence)

This is in CNF.

2. $\neg(p \wedge q) \leftrightarrow (p \vee q) \equiv (\neg(p \wedge q) \rightarrow (p \vee q)) \wedge ((p \vee q) \rightarrow \neg(p \wedge q))$
(Section 1.3, Table 8, Equivalence 1)

$\equiv (\neg\neg(p \wedge q) \vee (p \vee q)) \wedge (\neg(p \vee q) \vee \neg(p \wedge q))$ (Conditional-disjunction equivalence)

$\equiv ((p \wedge q) \vee (p \vee q)) \wedge (\neg(p \vee q) \vee \neg(p \wedge q))$ (Double-negation law)

$\equiv (((p \wedge q) \vee p) \vee q) \wedge (\neg(p \vee q) \vee \neg(p \wedge q))$ (1st Associative Law)

$\equiv ((p \vee (p \wedge q)) \vee q) \wedge (\neg(p \vee q) \vee \neg(p \wedge q))$ (1st Commutative Law)

$\equiv (p \vee q) \wedge (\neg(p \vee q) \vee \neg(p \wedge q))$ (1st Absorption Law)